

# Wael El-Dabah

## Security Research | Applied Cybersecurity & AI/ML

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### PROFILE

Software Engineering graduate with applied cybersecurity experimentation in network traffic analysis, machine-learning-assisted detection, threat intelligence, and vulnerability research. Experience in technical documentation, practical security assessments, and conveying results through teaching and presentations. Interested in research-oriented graduate study and security research roles.

### RESEARCH INTERESTS

Application security, vulnerability analysis, secure software engineering, network threat detection, AI/ML for cybersecurity and cyber threat intelligence.

### SELECTED RESEARCH & TECHNICAL PROJECTS

#### EyeGuard | AI-Driven Network Detection & Response | 2025 - Present

- Developed a passive NDR research prototype combining traffic analysis, anomaly detection and threat-intelligence context.
- Explored supervised and unsupervised approaches using Isolation Forest, LightGBM, Random Forest and XGBoost; experimented with CICIDS, NSL-KDD, UNSW-NB15 and CIC-IDS2017 benchmark datasets.
- Implemented packet capture and flow-feature processing with Python, Scapy and PyShark; connected enrichment sources including VirusTotal, AlienVault OTX, AbuseIPDB, Shodan and MXToolbox.
- Roadmap explores additional XDR-oriented correlation and policy-driven response; production deployment is not claimed.

#### Security Assessment / VAPT Project | Digital Egypt Pioneers Initiative

- Produced technical assessment documentation and reporting; public project material available on GitHub.

### RESEARCH & TECHNICAL METHODS

**Methods:** Experiment design, proof-of-concept implementation, dataset evaluation, network feature engineering, technical documentation, report writing.

**Technologies:** Python, Scapy, PyShark, Scikit-learn ecosystem, Isolation Forest, LightGBM, Random Forest, XGBoost, Git, Linux; Burp Suite, Nmap, Wireshark.

**Research communication:** Technical presentations, mentoring, stakeholder discussions, evidence-based findings.

### INDUSTRY, INSTRUCTION & MENTORING EXPERIENCE

#### Cyber Security Intern | IGRC Square | Jul 2025 - Dec 2025

- Supported web application penetration testing, mobile security testing, and network security assessments under senior supervision.
- Conducted vulnerability analysis and secure code reviews; documented reproducible evidence, impact, and remediation recommendations.
- Participated in technical discussions and communication of findings alongside senior consultants.

#### Cyber Security Instructor | Udacity | Dec 2024 - Mar 2025

- Delivered cybersecurity training, guided practical exercises, mentored learners, and evaluated technical progress.

#### Programming Instructor | Almentor | Sep 2024 - Dec 2024

- Taught Python programming through coding, debugging and structured problem-solving exercises.

### EDUCATION

B.Sc. in Software Engineering | Assiut National University, Faculty of Computers & Artificial Intelligence | 2022 - 2026

Overall grade: Very Good | Graduation project: EyeGuard (Excellent)

### CERTIFICATIONS & TRAINING

eJPT (Junior Penetration Tester) - Earned

HCIA-Security - Earned

Certified Ethical Hacker (CEH) preparation - NTI training

### AWARDS & ENGAGEMENT

4th nationwide - Digitopia Cybersecurity Competition (EyeGuard).

Best Team Leader - Digital Egypt Pioneers Initiative (DEPI); recognition by the Minister of Communications.

Cybersecurity instructor, Google Developer Student Club - delivered introductory security and penetration testing sessions.